

Training Configuration and Results Summary

Training Parameter	Value
Base Model	Llama 3.1 8B Instruct
Fine-tuning Method	LoRA (Low-Rank Adaptation)
Quantization	4-bit (BitsAndBytes)
Total Training Steps	717
Total Epochs	3.00
Best Model Checkpoint	Step 500
Best Validation Loss	0.2336
Training Examples	1,909
Validation Examples	213
LoRA Rank	8
LoRA Alpha	16
Learning Rate	2e-4
Batch Size	1
Gradient Accumulation Steps	8
Effective Batch Size	8
Max Sequence Length	1024
Optimizer	PagedAdamW-8bit